

Agile Test Process, Certifications, and Test Plan

Project: AutoFleet multi-robot supervision and V2X control stack

Program context: Master IISC Pro 2nd Year, Agile Project Management

Execution date: 2026-04-16

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1. Purpose

This document defines a standards-compliant Agile testing workflow for the AutoFleet project and records the tests that can be executed locally in the current software stack.

The goal is to make each implemented user story acceptable through explicit tests. Each story must have at least one acceptance condition, and each acceptance condition must be linked to one or more tests written in Behavior Driven Development form:

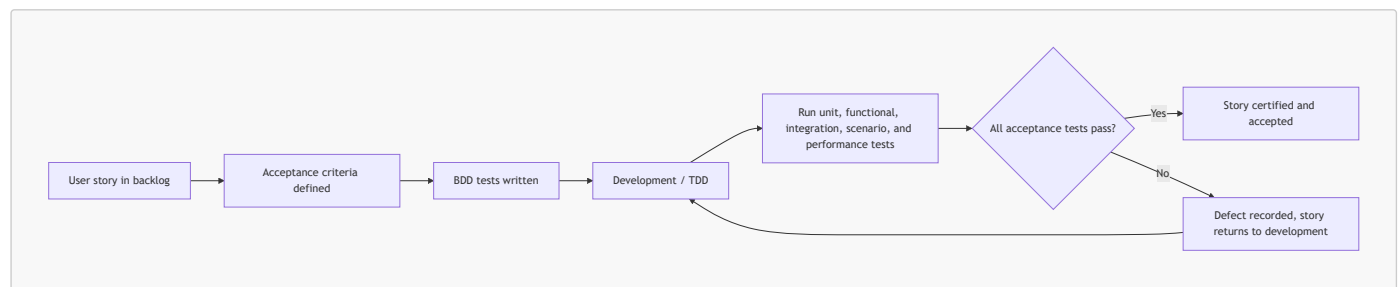
```
Given <initial system state>
When <event or user action>
Then <expected system state>
```

The tests in this document focus on the parts that can be validated in the current repository:

- Docker-based software stack startup
- Robot simulation telemetry
- V2X / MQTT command and acknowledgment flow
- Backend REST API
- Video worker stream registration and MJPEG proxy
- Perception, map, and coordination summaries
- Mission and formation orchestration
- Frontend and simulator monitor availability

Hardware-only items such as real LiDAR, motor encoders, Raspberry Pi 5 to ESP32 electrical communication, Xenomai real-time scheduling, and physical watchdog reset are listed as planned on-hardware validations. They are not certified by this local execution.

2. Agile Acceptance Workflow



Rules applied:

- A user story must have at least one acceptance condition.
- Each acceptance condition must be covered by one or more tests.
- Acceptance tests describe the expected execution of the story from the user or system point of view.
- If a story needs too many acceptance tests, the story is probably too large and should be split.
- Tests are not a final phase. They guide development and are rerun after each build to avoid regressions.

3. Test Plan Dimensions

Story Dimension

The test plan covers the main vertical slices available in the current stack:

- Robot telemetry from simulator to backend to frontend
- Command dispatch from frontend/API to backend to MQTT to simulator
- Video metadata from simulator to video worker to backend/frontend
- Perception and map summaries from video snapshots to backend/frontend
- Mission and formation APIs across backend state, MQTT commands, and robot acknowledgments

Test Type Dimension

Test type	Usage in this project
Unit test	Used for isolated hardware/software modules when available, for example motor encoder logic or sensor drivers. Not yet automated in this repository.
Functional test	Validates one subsystem behavior, for example simulator telemetry or video stream registration.
Integration test	Validates communication between components, for example backend, MQTT broker, robot simulator, video worker, and database.
Scenario test	Validates an operator-level workflow, for example mission start, return, and stop.
Performance test	Validates non-functional behavior, for example repeated V2X command traffic and latency.
Reliability / safety test	Validates fail-safe behavior such as watchdog reset. Planned for hardware validation.

Time Dimension

Moment	Test action
Before sprint development	Define acceptance criteria and BDD tests for each selected story.
During development	Add or refine tests when implementation details become clearer.
Before merging	Run all tests related to modified services and all regression smoke tests.
At sprint review	Present acceptance results and evidence to certify completed stories.
After every new build	Rerun automated tests to detect regressions.

Strategy Dimension

Responsibility	Role
Product owner / client	Validates acceptance criteria and story value.
Developer	Writes unit tests and supports automation.
Tester / integrator	Defines BDD acceptance tests and executes scenario tests.
Team	Reviews failures and decides whether the story is accepted or returned to development.

If a test is KO:

1. Record the failed test ID, environment, input data, and observed result.
2. Create or update a defect task.
3. Return the story to development if the failure blocks acceptance.
4. Rerun the failed test and all related regression tests after the fix.

Environment Dimension

Executed environment:

Component	Local endpoint / service
Docker Compose file	infra/compose.yml
Frontend	http://127.0.0.1:3000
Simulator monitor	http://127.0.0.1:3000/sim-monitor.html
Backend API	http://127.0.0.1:8200/api/v1
Video worker	http://127.0.0.1:8400
MQTT broker	127.0.0.1:3889
PostgreSQL	127.0.0.1:5432
Robot simulator	Docker service robot-sim , robots R1 , R2 , R3

Evidence files:

- [docs/test-evidence/api/acceptance-results-20260416-102943.json](#)
- [docs/test-evidence/api/v2x-load-results-20260416-103551.json](#)

Screenshot placeholders are intentionally left empty because screenshots will be added manually later.

4. Data-Driven Test Figures

The following figures were generated directly from the JSON evidence files with [tools/generate_test_figures.py](#). They are designed for report or paper usage: high-resolution PNG, consistent typography, explicit acceptance thresholds, and captions tied to the test evidence.

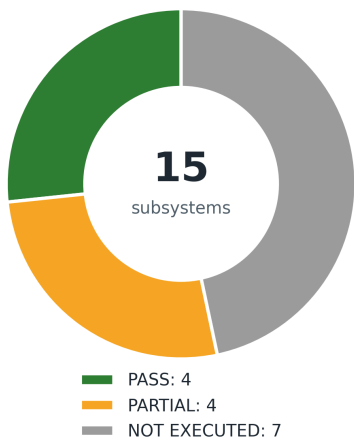
Executed Acceptance Test Certification Matrix

T01	Docker stack	PASS Integration
T02	Robot simulation	PASS Functional
T03	Video worker	PASS Functional / integration
T04	Perception + map	PASS Functional
T05	V2X command ACK	PASS Integration
T06	Formation	PASS Integration
T07	Mission scenario	PASS Scenario
T08	Frontend pages	PASS Functional
T09	V2X light load	PASS Performance
T10	Persistence + events	PASS Integration

Figure 1. Local MVP acceptance tests all reached their configured acceptance criteria. Certified

Figure 1 shows that the ten tests executed in the local MVP scope reached their acceptance criteria.

Subsystem validation status



Mobile robot simulation	PASS
LiDAR simulation	NOT EXECUTED
Actuators / motors	NOT EXECUTED
Motor encoder	NOT EXECUTED
Raspberry Pi 5 - ESP32	NOT EXECUTED
Embedded architecture	PARTIAL
System watchdog	NOT EXECUTED
Ultrasonic sensor	NOT EXECUTED
Embedded camera	PARTIAL
SLAM	NOT EXECUTED
Autonomous navigation	PARTIAL
Multi-sensor integration	PARTIAL
V2X communication	PASS
V2X performance	PASS
Complete software integration	PASS

Figure 2. Certification is intentionally limited to the software/simulation scope available in the repository.

Figure 2 separates certified software/simulation scope from partial or not-yet-executed hardware validation.

Runtime Evidence Coverage

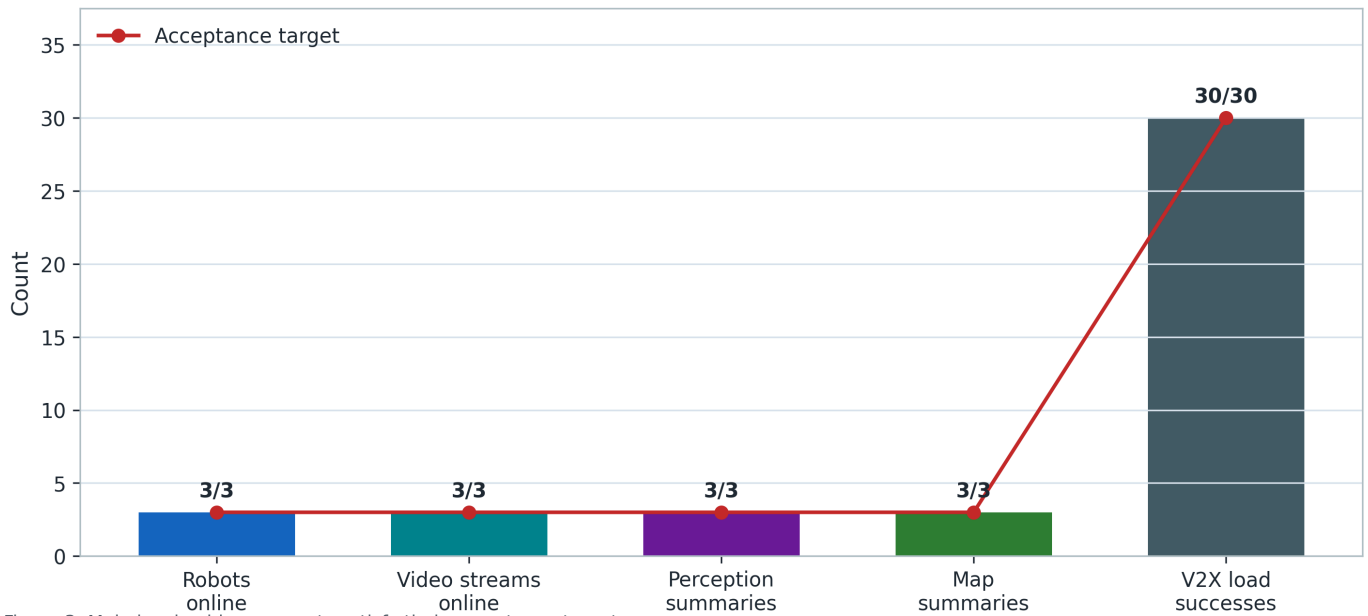


Figure 3. Main local evidence counts satisfy their acceptance targets.

Figure 3 summarizes the main evidence counts: robots online, video streams online, perception summaries, map summaries, and successful V2X load requests.

V2X Teleoperation Response Time Under Light Load

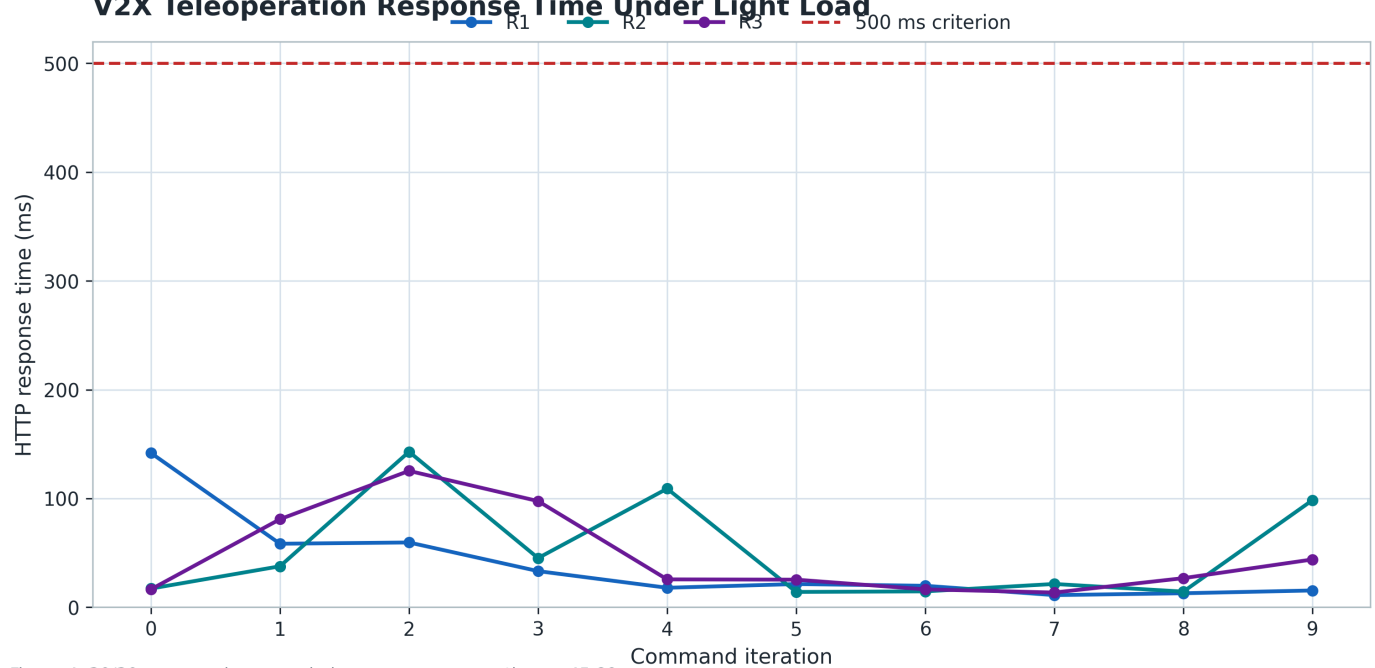


Figure 4. 30/30 commands succeeded; average response time = 45.89 ms.

Figure 4 shows command response time over ten iterations for each robot. The red dashed line is the 500 ms performance criterion.

V2X Latency Distribution by Robot

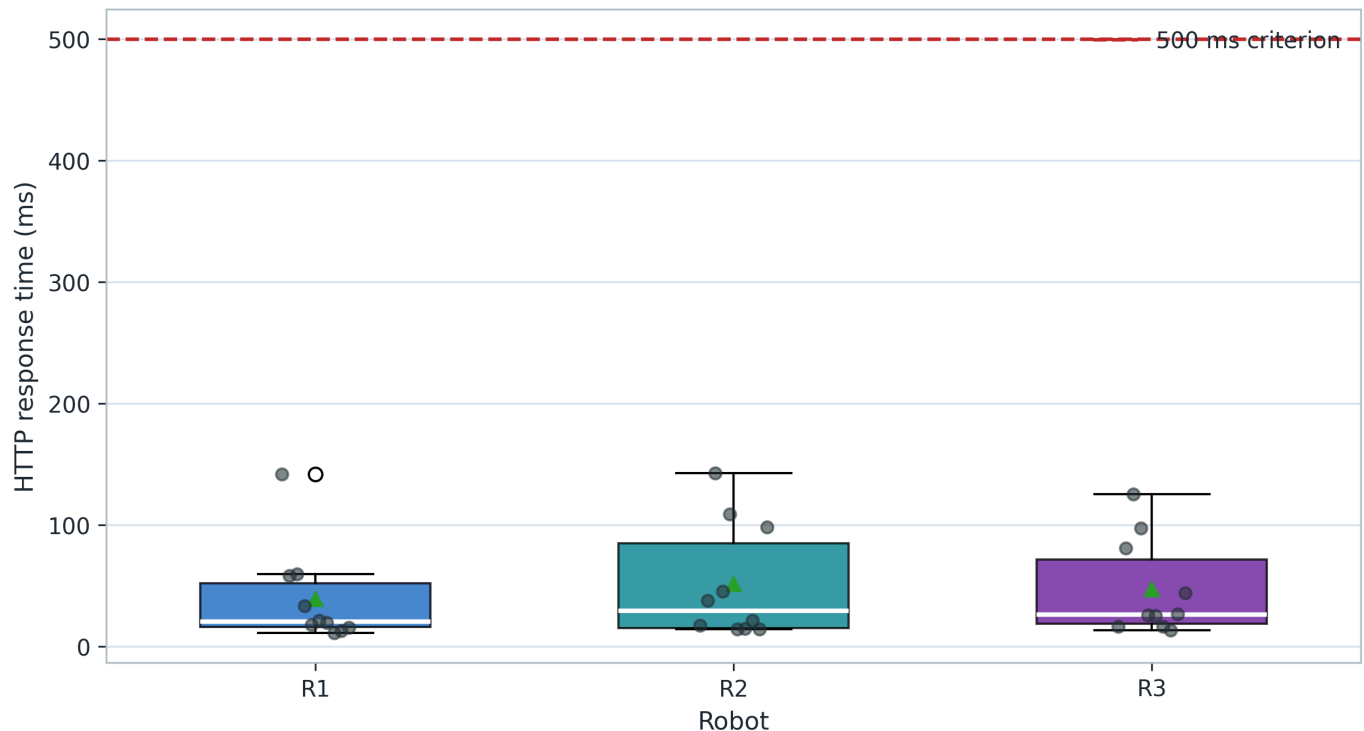


Figure 5. Per-robot latency stayed far below the configured 500 ms acceptance threshold.

Figure 5 compares latency distributions for R1, R2, and R3; all observations remain below the acceptance threshold.

Robot Network Telemetry Snapshot Before Command Tests

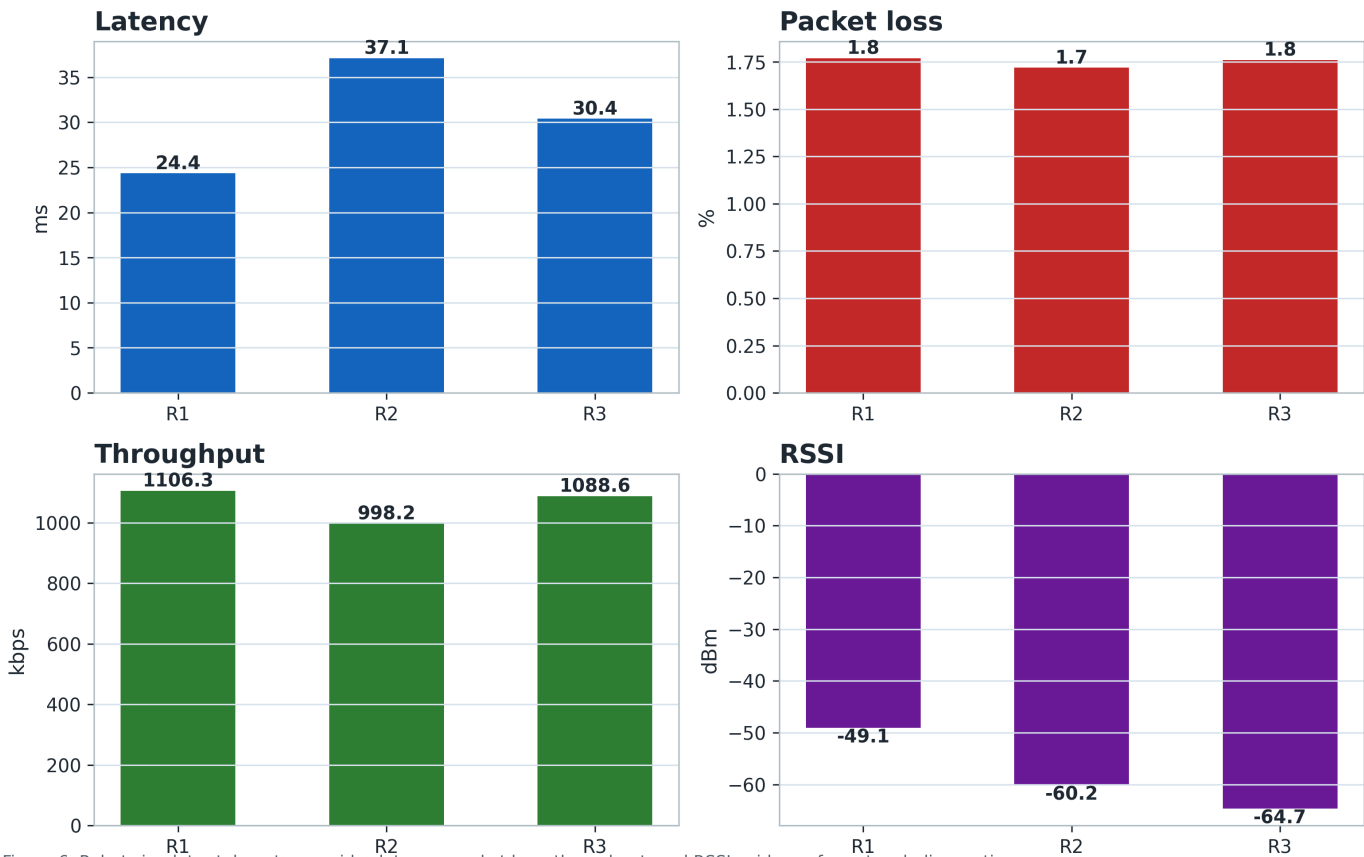


Figure 6. Robot simulator telemetry provides latency, packet loss, throughput, and RSSI evidence for network diagnostics.

Figure 6 uses robot telemetry to compare latency, packet loss, throughput, and RSSI before command tests.

Agile Story Acceptance Pipeline



KO path: record defect -> fix implementation -> rerun failed and regression tests

Figure 7. Tests are part of story definition and acceptance, not a final waterfall phase.

Figure 7 provides a report-ready version of the Agile story acceptance pipeline.

5. Executed Acceptance Tests

ID	Subsystem	Objective	Type	Given / When / Then	Acceptance criterion	Result	Evidence
T01	Docker stack	Validate that the complete software stack starts locally	Integration	Given Docker Desktop is running and <code>infra/compose.yml</code> is available. When <code>docker compose up -d</code> starts the stack. Then backend, MQTT, database, workers, simulator, and frontend are reachable.	Backend health is <code>ok</code> , MQTT connected, database available.	PASS	acceptance-results-20260416-102943.json
T02	Robot simulation	Validate simulated robot telemetry	Functional	Given <code>robot-sim</code> publishes robots <code>R1</code> , <code>R2</code> , <code>R3</code> . When <code>/api/v1/robots</code> is requested. Then three robots are listed with fresh telemetry.	<code>robots_online_count = 3</code> .	PASS	acceptance-results-20260416-102943.json
T03	Video worker	Validate simulated video source ingestion	Functional / Integration	Given each robot publishes <code>file:///artifacts/demo/highway-forward.mp4</code> . When <code>/streams</code> is requested from the video worker. Then three MJPEG proxy streams are registered.	<code>video_streams_online_count = 3</code> .	PASS	acceptance-results-20260416-102943.json
T04	Perception and map summaries	Validate software perception outputs from snapshots	Functional	Given the video worker provides snapshots. When the perception worker analyzes them. Then backend exposes perception and map summaries.	<code>perception_count = 3</code> , <code>map_summary_count = 3</code> .	PASS	acceptance-results-20260416-102943.json
T05	V2X command ACK	Validate command dispatch and acknowledgment	Integration	Given robot <code>R1</code> is online. When a teleop command and a STOP command are sent through the backend. Then the simulator receives the commands and publishes ACK messages.	ACK status is <code>ACCEPTED</code> for teleop and STOP.	PASS	acceptance-results-20260416-102943.json
T06	Formation orchestration	Validate leader/follower state management	Integration	Given robots <code>R1</code> , <code>R2</code> , <code>R3</code> are online. When follow formation starts with <code>R1</code> as leader and <code>R2</code> , <code>R3</code> as followers. Then backend formation state becomes enabled.	<code>formation_enabled_during_test = true</code> .	PASS	acceptance-results-20260416-102943.json
T07	Mission scenario	Validate mission start, return, and stop workflow	Scenario	Given a mission <code>qa-20260416-102943</code> with three robots. When the mission is started, returned, and stopped. Then the mission state is stored and reaches a final stopped state.	<code>mission_final_status = STOPPED</code> .	PASS	acceptance-results-20260416-102943.json
T08	Frontend pages	Validate operator UI availability	Functional	Given the frontend container is running. When <code>/</code> and <code>/sim-monitor.html</code> are requested. Then both pages return HTTP 200.	<code>frontend_ok = true</code> , <code>sim_monitor_ok = true</code> .	PASS	acceptance-results-20260416-102943.json

ID	Subsystem	Objective	Type	Given / When / Then	Acceptance criterion	Result	Evidence
T09	V2X light load	Validate basic V2X stability under repeated commands	Performance	Given three robots are online. When 30 teleop commands are sent across R1, R2, and R3. Then all requests complete and the protocol has no pending commands after reset.	30/30 successful requests, 0 failures, average HTTP response 45.89 ms, pending commands 0, all robots online.	PASS	v2x-load-results-20260416-103551.json
T10	Persistence and events	Validate runtime persistence availability	Integration	Given PostgreSQL is configured. When backend health and events are requested. Then database is reported available and runtime events are exposed.	database_available = true; events endpoint responds.	PASS	acceptance-results-20260416-102943.json

6. Certification Summary

User story / capability	Certification status	Basis
As an operator, I can see the state of the simulated fleet.	Certified	T01, T02, T08
As an operator, I can send teleoperation commands to a robot.	Certified	T05
As an operator, I can start and stop a mission for multiple robots.	Certified	T07
As an operator, I can enable and disable leader/follower formation.	Certified	T06
As the system, I can register and proxy simulated robot video streams.	Certified	T03
As the system, I can publish perception and map summaries from video snapshots.	Certified	T04
As the system, I can support light V2X traffic across three robots.	Certified for local light-load simulation	T09
As the system, I can control real motors through Raspberry Pi 5 and ESP32.	Not certified yet	Requires hardware bench test
As the system, I can localize and map with ROS2 SLAM and LiDAR.	Not certified yet	Requires ROS2 / LiDAR simulation or real sensor environment
As the system, I can guarantee real-time scheduling under Xenomai.	Not certified yet	Requires target real-time OS environment
As the system, I can recover through hardware/software watchdog.	Not certified yet	Requires fault injection on the embedded target

7. Mapping to the Subsystem Test Matrix

Subsystem	Local validation status	Comment
Mobile robot simulation	PASS	Validated through robot-sim telemetry and command response.
LiDAR in simulation	NOT EXECUTED	Current repository does not expose a ROS2 /scan LiDAR simulator.
Actuators / motors	NOT EXECUTED	Requires physical motor bench or ESP32 motor controller simulator.
Motor encoder	NOT EXECUTED	Requires encoder input data or hardware bench.
Raspberry Pi 5 to ESP32 communication	NOT EXECUTED	Requires serial/network link to ESP32 or a dedicated mock.
Embedded control architecture	PARTIAL	Software-level command separation is validated through backend to simulator; hardware role separation is not certified.
System watchdog	NOT EXECUTED	Requires controlled software lockup or target watchdog configuration.
Ultrasonic sensor	NOT EXECUTED	Requires physical sensor or sensor simulator.
Embedded camera	PARTIAL	File-based simulated camera stream is validated; real camera capture is not certified.
SLAM localization and mapping	NOT EXECUTED	Current stack exposes map summaries from perception heuristics, not ROS2 SLAM.
Autonomous navigation	PARTIAL	Mission orchestration is validated; path planning and collision-free physical navigation are not certified.
Multi-sensor integration	PARTIAL	Video, perception, telemetry, and coordination run together; physical LiDAR/camera/ultrasound fusion is not certified.
V2X communication	PASS	Backend, MQTT, robot simulator, ACKs, and telemetry are validated.
V2X performance	PASS for light load	30 commands, 0 failures, average HTTP response 45.89 ms.

Subsystem	Local validation status	Comment
Complete system integration	PASS for local MVP stack	Full Docker software stack validated; hardware integration remains future work.

8. Screenshot Placeholders

Screenshots should be added manually at these points:

- 1. Docker container status showing all services running.
- 2. Backend health endpoint: `http://127.0.0.1:8200/api/v1/health`.
- 3. Robot list endpoint: `http://127.0.0.1:8200/api/v1/robots`.
- 4. Video stream registry: `http://127.0.0.1:8400/streams`.
- 5. Main dashboard: `http://127.0.0.1:3000`.
- 6. Simulator monitor: `http://127.0.0.1:3000/sim-monitor.html`.
- 7. Mission final state for the executed QA mission.
- 8. V2X load result JSON or terminal output.

Recommended image naming:

```
docs/test-evidence/screenshots/01-docker-compose-ps.png
docs/test-evidence/screenshots/02-backend-health.png
docs/test-evidence/screenshots/03-robots-api.png
docs/test-evidence/screenshots/04-video-streams.png
docs/test-evidence/screenshots/05-dashboard.png
docs/test-evidence/screenshots/06-sim-monitor.png
docs/test-evidence/screenshots/07-mission-final-state.png
docs/test-evidence/screenshots/08-v2x-load-result.png
```

9. Reproduction Commands

Start the full local stack:

```
cd infra
$env:AUTOFLEET_API_PORT='8200'
$env:AUTOFLEET_PUBLIC_HOST='127.0.0.1'
docker compose up -d
```

Check service health:

```
Invoke-RestMethod http://127.0.0.1:8200/api/v1/health
Invoke-RestMethod http://127.0.0.1:8200/api/v1/robots
Invoke-RestMethod http://127.0.0.1:8400/streams
```

Send a teleop command:

```
Invoke-RestMethod `
-Method POST `
-ContentType 'application/json' `
-Uri http://127.0.0.1:8200/api/v1/teleop/R1 `
-Body '{"linear_x":0.35,"angular_z":0.12,"ttl_ms":700}'
```

Reset a robot to safe state:

```
Invoke-RestMethod `
-Method POST `
-ContentType 'application/json' `
-Uri http://127.0.0.1:8200/api/v1/robots/R1/command `
-Body '{"type":"STOP","args":{"reason":"manual-reset"},"ttl_ms":2000}'
```

Generate the figures from saved evidence:

```
python tools\generate_test_figures.py
```


10. Acceptance Decision

The local MVP software stack is accepted for the tested scope.

Certified scope:

- Local Docker startup
- Simulated three-robot telemetry
- Backend REST API health
- MQTT/V2X command and ACK loop
- Video worker stream ingestion from local demo video
- Perception and map summaries
- Mission and formation API workflows
- Frontend page availability
- Light V2X command-load stability

Not certified in this execution:

- Physical motors and actuators
- Motor encoders
- Raspberry Pi 5 to ESP32 hardware communication
- Real LiDAR and ultrasonic sensors
- ROS2 SLAM / RViz2 visualization
- Xenomai real-time scheduling
- Hardware watchdog reset
- Physical autonomous navigation without collision